Application Or website

Application architect

Architect ka role kya hota hai?

Architect decide karta hai:

Application ka structure kaisa hoga (1-tier, 3-tier ya microservices)

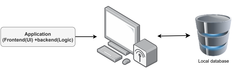
Kaun sa technology use hoga (Java, .NET, Node, etc.)

Kaun sa cloud (Azure / AWS / GCP)

Security kaise implement hogi

Scaling kaise hoga (10 user vs 10 lakh user)

1-Tier Architecture (Single Layer)



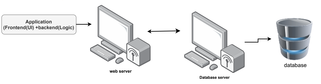
Sab kuch ek hi machine me hota hai

UI (Frontend),Business Logic,Database

Example:MS Excel,Local desktop app

Never:
Scalability nahi hoti
Multi-user support weak

2-Tier Architecture (Client-Server))



User → Client App → Database

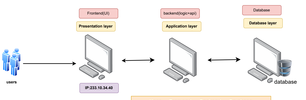
Do layer hoti hain:

Client (UI + logic),Database Server

Example:Direct DB connect apps (like old banking apps),Inventory Management

Security risk (client directly DB se connect)
Hard to scale

3-Tier Architecture (Most Common 🔥)



User → Frontend → Backend → Database

3 layers hoti hain:

Presentation Layer (Frontend – Browser, React, Angular)

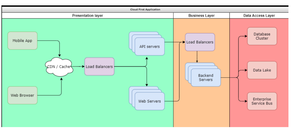
Application Layer (Backend – API, Node.js, Java, .NET)

Data Layer (Database – SQL, MongoDB)

Example:amazon,netflix

✅ Benefit:Secure,Scalable,Easy maintenance

4-Tier (🔥 3-tier + extra layer add hoti hai)



4 layer hoti hain

✅ 4 layers hoti hain:

Client (Browser / App)

Web Server (Nginx / Apache) 🔥

Application Server (Backend / API)

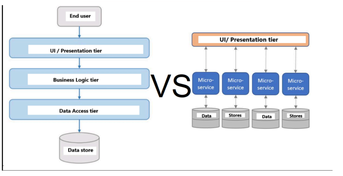
Database

Example:Enterprise apps,Large scale systems.

✅ Benefit:Load distribution,Better performance

User → Web Server → App Server → DB

N-Tier Architecture (Advanced / Real World 🔥)



⚠ N = Many layers

Load Balancer

API Gateway

Microservices

Cache (Redis)

Message Queue (Kafka)

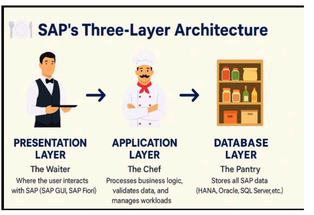
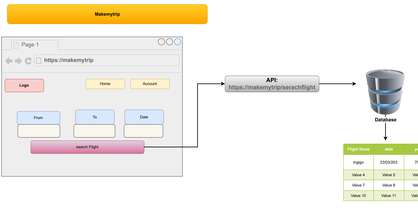
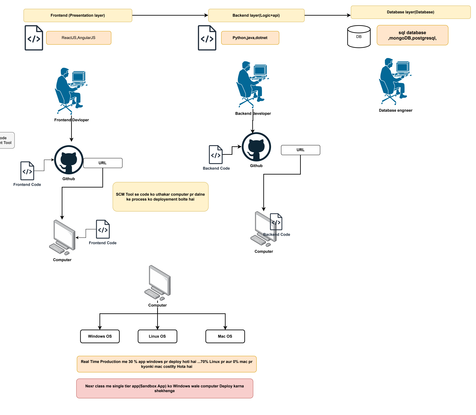
Database

✅ Example:Netflix,Amazon,Swiggy

Benefit:High scalability,Fault tolerance,Microservices support

User → LB → API Gateway → Services → DB

Cloud vs On-premise

Monolithic Infra → Virtualization Technology

3-Tier Architecture (Most Common 🔥)